*Part IV — Rotation, Equilibrium, and Materials · Chapter 21*

# Center of Mass and Stability

*How the distribution of mass within an object determines balance, tipping, and stability*

*Why does a racing car sit so low to the ground? Why do sumo wrestlers adopt a wide, crouching stance? Why does a loaded truck tip over on a curve that an empty one takes safely? Why must a crane extend its counterweight before lifting a heavy load? All of these questions have the same physical answer — one that goes deeper than simply 'it's heavy.' The answer lies in where the mass is located, and how that location relates to the object's base of support.*

## 21.1 What the Center of Mass Represents

Every object is made up of an enormous number of particles — atoms and molecules — each with its own mass and position. Analyzing each particle individually is impossible in practice. The center of mass is a powerful simplification: it is a single point that represents the average position of all the mass in a system. For most purposes, gravity, inertia, and the overall translational motion of an extended object can be analyzed by treating all the mass as if it were concentrated at this one point.

More precisely, the center of mass is the mass-weighted average position of all the particles in a system. Particles with greater mass pull the center of mass closer to themselves; particles with lesser mass have less influence. The center of mass of a symmetrical, uniformly dense object lies at its geometric center. But for irregular objects, or objects made of different materials, the center of mass can be far from the geometric center — and sometimes outside the physical object altogether.

The center of mass has a remarkable property: when external forces act on a system, the center of mass accelerates exactly as if all the mass were concentrated there and all the forces applied to it. This is why we can treat a thrown baseball as a point particle moving in a parabolic arc, even though it is actually rotating and vibrating — its center of mass follows the ideal projectile path.

## 21.2 Center of Mass vs. Geometric Center

The geometric center (or centroid) of an object is the average position of its boundary — the central point of its shape. For a uniform solid, the center of mass and geometric center coincide. But the two diverge whenever density varies across the object.

A hammer provides a clear example. Its handle is long and light; its head is short and heavy.

The geometric center of the hammer lies roughly in the middle of its length, in the handle. The center of mass lies much closer to the head, where most of the mass is concentrated. If you balance a hammer on your finger, the balance point — the center of mass — is near the head, not the middle of the handle.

Similarly, the human body's center of mass shifts constantly with posture. Standing upright, it lies roughly at hip level, about 57–60% of a person's height from the ground. When you extend your arms overhead, raise a leg, or bend over, the center of mass shifts accordingly — and can actually move outside the body entirely in extreme postures.

This distinction matters enormously in engineering. A steel I-beam has a geometric center at its midpoint, but if one end is encased in concrete, the center of mass of the composite system shifts toward the heavier concrete end. Designers must locate the true center of mass, not the geometric centroid, when analyzing balance and stability.

## 21.3 Locating the Center of Mass

For a system of discrete objects (separate masses at known positions), the center of mass can be found by computing the weighted average position. In one dimension, if masses $m_1$, $m_2$, $m_3$, ... are located at positions $x_1$, $x_2$, $x_3$, ... along a line, the center of mass position is:

$$x_{cm} = (m_1x_1 + m_2x_2 + m_3x_3 + \ldots) / (m_1 + m_2 + m_3 + \ldots)$$

$$x_{cm} = \Sigma(m_i x_i) / \Sigma m_i$$

The same formula applies in the y-direction to find $y_{cm}$. In three dimensions, all three coordinates are calculated independently. For uniform geometric shapes — rectangles, circles, triangles, and combinations thereof — the center of mass of each piece lies at its geometric center, and the pieces can then be combined using the formula above.

For continuous objects with non-uniform density, calculus methods are required — integration replaces the sum. At this level, we will work with combinations of simple uniform shapes, which is sufficient for most engineering and physics problems.

Practical method for irregular shapes: suspend the object from one point and let it hang freely. The center of mass will lie somewhere along the vertical line through the suspension point (because only a line through the center of mass produces no net torque). Mark this line on the object. Suspend from a second point and mark again. The intersection of the two lines is the center of mass.

## 21.4 Center of Gravity and Uniform Gravitational Fields

The center of gravity is the point where the entire weight of an object may be considered to act for the purposes of calculating torques due to gravity. In a uniform gravitational field — one where g is the same everywhere on the object — the center of gravity and the center of mass are exactly the same point. Since the gravitational field near Earth's surface is essentially uniform for any ordinary object, center of mass and center of gravity are effectively synonymous in introductory physics.

The distinction matters only for very large objects (such as a spacecraft spanning a significant fraction of the distance to the Moon) where gravity varies measurably from one end to the other. For all problems in this book, center of mass and center of gravity can be used interchangeably.

This equivalence justifies a common simplification: when drawing free-body diagrams, we draw the weight of an extended object as a single downward arrow from its center of mass. This correctly represents the net gravitational torque about any external pivot.

> **Bridge Note** *In general relativity, gravity is not a force in the traditional sense — it is curvature of spacetime. The concept of 'center of gravity' becomes more subtle. For introductory purposes, Newtonian gravity with center of mass and center of gravity as identical is entirely adequate.*

## 21.5 Why Stability Depends on Center of Mass

An object is stable if, when displaced slightly from its equilibrium position, restoring forces or torques bring it back. It is unstable if a small displacement causes forces or torques that push it further away from equilibrium. The position of the center of mass is the central factor determining which of these applies.

Consider an object resting on a surface. Gravity acts downward through the center of mass. The surface provides an upward normal force through the base of support. As long as the vertical line through the center of mass passes through the base of support, the gravitational torque tends to restore the object when it tilts slightly — pushing it back toward upright.

If the object tilts enough that the center of mass moves outside the base of support, the gravitational torque now acts to rotate the object further away from upright — it tips over. The critical condition is precisely the moment when the vertical line through the center of mass passes through the edge of the base: this is the tipping point. Any further tilt beyond this results in overturning.

Lower center of mass means the critical tipping angle is larger — the object can lean more before tipping. A wider base of support means the center of mass must travel a greater horizontal distance before reaching the edge — it also increases stability. Both factors work together, and good design exploits both.

## 21.6 Base of Support and Stability

The base of support is the area enclosed by the outermost contact points between an object and the surface it rests on. For a four-legged chair, the base of support is the rectangle (or polygon) formed by the four legs. For a person standing with feet shoulder-width apart, it is the roughly rectangular area between and around the feet. For a person standing on one foot, it is the small area of the single foot in contact with the ground.

Stability increases as the base of support increases. This is why tripods, three-legged stools, and four-legged tables are stable: they have a well-defined, reasonably large base of support. It is why a box is more stable lying flat (large base) than standing on end (small base). It is why a person balancing on tiptoe is unstable: the base of support is tiny.

For an object to be in stable equilibrium, the vertical projection of the center of mass must fall within the base of support. This is not merely a rule of thumb — it follows directly from the torque analysis. When the center of mass is above a point within the base, any small tilt creates a restoring torque. When the center of mass is above a point outside the base, the torque causes continued tipping.

| Configuration | Base of Support | Stability |
|---|---|---|
| Book lying flat | Large rectangle | Very stable |
| Book standing on spine | Narrow rectangle | Moderately stable |
| Book balanced on corner | Point (corner) | Highly unstable |
| Person: feet wide apart | Wide polygon | Very stable |
| Person: feet together | Narrow strip | Less stable |
| Person: on one foot | Single foot area | Requires active balance |

*Table 21.1 How base of support area determines static stability.*

## 21.7 Tipping and Rotational Instability

Tipping occurs when a tilting object's center of mass reaches the tipping point — the vertical line directly above the pivot edge of the base. At that exact moment, the net gravitational torque is zero about the pivot edge (the center of mass is directly above it). Any further tilt beyond this critical angle sends the center of mass outside the base, producing a torque that accelerates tipping. The object then falls.

The critical tipping angle can be estimated geometrically. For a rectangular block of height h and base width b, the center of mass is at height h/2 and horizontal distance b/2 from the edge. The critical angle is arctan(b/h). A tall, narrow block (small b, large h) has a small critical angle — it tips easily. A wide, flat block (large b, small h) has a large critical angle — it is very resistant to tipping.

This geometry explains many everyday phenomena. A filing cabinet filled only at the top is dangerously unstable because the effective center of mass is high. Manufacturers specify maximum drawer extension limits for this reason. A double-decker bus is engineered with heavy components low in the chassis so that the center of mass stays low even when the upper deck is full.

Vehicles are tested for rollover resistance by measuring the height of the center of mass relative to the track width (distance between wheels). Sports utility vehicles (SUVs) became notorious in the 1990s for relatively high centers of mass combined with narrow track widths — a combination that produced low critical tipping angles and increased rollover risk during sharp turns.

## 21.8 Shifting Center of Mass in Motion

The center of mass of a system is not fixed — it changes whenever mass is redistributed. For a rigid object, the center of mass is fixed relative to the object. But for a system of objects, or for a flexible object (like a human body), the center of mass shifts as parts move.

When a person bends forward, the center of mass shifts forward and downward — if they bend far enough, the center of mass can move outside the body and in front of the feet. Without compensating movements (like extending the arms backward or bending the knees), they would fall forward. The body's balance system continuously monitors center of mass position and makes constant small adjustments to keep it over the base of support.

In projectile motion, even a spinning or tumbling object's center of mass follows a smooth parabola — while the rest of the object moves in complex patterns around that path. This is famously illustrated by a rotating wrench thrown through the air: the wrench spins and tumbles, but the geometric center traces an irregular path while the center of mass traces the clean parabola.

Astronauts, divers, gymnasts, and ballet dancers all manipulate their body's center of mass deliberately: pulling in during a spin to speed up rotation, extending to slow down, shifting weight to change trajectory. Understanding center of mass is central to any serious analysis of human movement.

## 21.9 Stability in Engineering and Architecture

Engineers must design structures that remain stable under every expected loading condition — including wind, earthquake, snow, and dynamic loads from moving vehicles or machinery. Stability analysis is one of the most fundamental and consequential tasks in structural engineering.

The principle of keeping the center of mass low and the base of support wide appears throughout structural design. Pyramids are inherently stable — massive base, center of mass very low. The ancient Egyptians may not have known the physics, but they arrived at the same optimal form by trial and error over millennia. Modern high-rise buildings use foundations that spread load broadly and design mass distributions that keep the effective center of mass well below the critical height.

Cranes pose a specific challenge: the load being lifted is far from the crane's base and creates a large torque about the crane's tipping edge. Mobile cranes deploy outriggers — extendable supports — to widen the effective base of support before lifting. Tower cranes attach their base to the building structure to add extra restraint. Counterweights on the opposite end of the crane boom shift the center of mass back over the base.

Bridges must remain stable not just under their own weight but under variable live loads — vehicles that can be on one side but not the other. The design of every bridge includes analysis of the worst-case load distribution, ensuring that the structure never reaches a configuration where overturning or collapse becomes possible.

Retaining walls, dams, and sea walls are particularly susceptible to overturning because water pressure creates large horizontal forces with long moment arms. Their design requires careful analysis of where the resultant force acts relative to the base, and whether it stays within a safe central zone that prevents tipping.

## 21.10 Stability in Sports and Biomechanics

Athletic performance depends critically on managing the body's center of mass. The principles of stability, tipping, and base of support directly translate into coaching cues, technique refinements, and injury prevention strategies.

A defensive stance in wrestling, football, or basketball involves a wide stance and bent knees — both lower the center of mass and widen the base of support, making the athlete harder to knock over. This is not arbitrary convention; it is optimal physics. A narrow, upright stance has a high center of mass and small base: easy to unbalance. A low, wide stance has a low center of mass and large base: much more resistant to displacement.

Sprinters at the start of a race intentionally lean forward so far that their center of mass is ahead of the base of support. This is controlled instability — they are on the verge of falling forward, which drives them to accelerate rapidly. The starting blocks provide the force to push them forward before they can fall. Once at top speed, the body's center of mass is again held over the support foot at each stride.

Rock climbers must constantly keep their center of mass over their foothold. When the center of mass drifts outside the contact point, they fall — not because of insufficient grip strength, but because of torque. Experienced climbers develop an intuitive feel for center of mass management, moving their hips and body weight deliberately to maintain balance on tiny holds.

In physical therapy, gait analysis measures the path of a patient's center of mass during walking.

Deviations from a healthy path indicate muscle weakness, joint problems, or neurological issues. Prosthetic limb design is optimized to reproduce the center of mass trajectory of a natural limb, making the gait as smooth and energy-efficient as possible.

## 21.11 Suspended and Hanging Objects

When an object hangs freely from a single suspension point, it will always orient itself so that its center of mass is directly below the point of suspension. This is because any other orientation would create a net gravitational torque that rotates the object until the center of mass reaches the lowest possible point — directly below the pivot.

This behavior provides a simple experimental method for finding the center of mass of any irregular flat object. Hang the object from one corner and draw a vertical line from the suspension point through the object. The center of mass lies on this line. Hang it from a different point and draw another line. The intersection of the two lines is the center of mass. This works because the object always rotates until its center of mass is directly below the suspension point, so the vertical line through any suspension point must pass through the center of mass.

This principle is used in product design and manufacturing. Aircraft and spacecraft components must have precisely determined center of mass positions; if the center of mass of a rocket payload is off-axis, it causes unplanned torques during flight. Medical devices such as prosthetics must have centers of mass calibrated for natural movement.

For a hanging object in equilibrium under multiple cables or ropes, the same torque analysis used in Chapter 20 applies. Each cable creates a torque about any chosen pivot, and for equilibrium, the net torque must be zero. The center of mass — where gravity effectively acts — is the key reference point for these calculations.

## 21.12 Common Mistakes in Stability Analysis

Stability problems are conceptually rich but also prone to specific misunderstandings. Recognizing these pitfalls prevents errors on both problem sets and in real-world applications.

- Confusing center of mass with geometric center. Always check whether the object is uniform. Non-uniform objects have their center of mass shifted toward the region of greater density or mass.
- Forgetting that center of mass can lie outside the physical object. A ring, a boomerang, a U-shaped object — all have centers of mass in the empty space in the middle. This does not make the concept wrong; it just means the effective point of gravitational action is outside the material.

- Identifying the base of support incorrectly. The base of support is the polygon formed by the outermost contact points, not just the areas touching the ground. For a person, it includes the region between and around the feet, not just the foot prints themselves.
- Assuming that a lower center of mass always means greater stability. True in general, but the base of support matters too. A very low center of mass on a tiny base can still be unstable if a small displacement moves the center of mass outside the base.
- Confusing static stability with dynamic stability. An object can be statically stable (in equilibrium when stationary) but dynamically unstable (small oscillations grow). Conversely, bicycles are statically unstable but dynamically stable when moving. These are distinct concepts.
- Treating the center of mass as fixed for all postures. For flexible objects and human bodies, the center of mass changes with configuration. Always re-evaluate its position for the specific posture or configuration you are analyzing.

# Chapter Summary

The center of mass is the mass-weighted average position of all particles in an object or system. It is the point through which gravity effectively acts and the point that follows Newton's laws of motion for the system as a whole.

The center of mass and geometric center coincide only for uniform objects with symmetric shapes. For non-uniform objects, the center of mass shifts toward the region of greater mass density. It can lie outside the physical object entirely (rings, arches, horseshoe shapes).

In a uniform gravitational field, center of mass and center of gravity are identical. Near Earth's surface, they can always be treated as the same point.

An object is stable when the vertical projection of its center of mass falls within the base of support. If the center of mass moves outside the base, the gravitational torque causes tipping rather than restoring.

Stability increases with: (1) lower center of mass, and (2) wider base of support. Both factors increase the critical tipping angle.

The tipping point is reached when the center of mass is directly above the pivot edge of the base. Beyond this point, the object falls; short of it, it recovers.

For hanging objects, the center of mass always comes to rest directly below the suspension point. This allows the center of mass to be located experimentally.

Center of mass principles explain vehicle rollover, athletic stance, crane counterweighting, structural design, and the biomechanics of human movement.

## Key Terms

| Term | Definition |
|---|---|
| **Center of mass** | The mass-weighted average position of all particles in an object or system; the point through which gravity effectively acts on an extended object. |
| **Geometric center (centroid)** | The average position of an object's boundary or shape; coincides with the center of mass only for uniform objects. |
| **Center of gravity** | The point where the net gravitational torque on an object can be considered to act; identical to the center of mass in a uniform gravitational field. |
| **Base of support** | The region enclosed by the outermost contact points between an |

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|---|---|
| | object and the surface it rests on; the object is stable only when the center of mass projects vertically into this region. |
| **Static stability** | The property of returning to equilibrium after a small displacement; requires that the center of mass projection remain within the base of support. |
| **Tipping point** | The critical angle at which the center of mass lies directly above the pivot edge; beyond this, the object overturns. |
| **Restoring torque** | A torque that acts to return an object to its equilibrium position when displaced. |
| **Overturning torque** | A torque that acts to rotate an object further away from its equilibrium position, causing it to tip or fall. |
| **Stable equilibrium** | Equilibrium in which a displaced object experiences a restoring force or torque that returns it to the original position. |
| **Unstable equilibrium** | Equilibrium in which a displaced object experiences a force or torque that moves it further from the original position. |
| **Neutral equilibrium** | Equilibrium in which displacement produces no net restoring or overturning force or torque (e.g., a ball on a flat surface). |

## Concept Review Questions

Answer the following in complete sentences. No formulas are required.

1. Explain in your own words what the center of mass represents. Why is it useful to define such a point?
2. A hammer and a pencil are both uniform cylinders of similar length. Which has its center of mass closer to the geometric center? Explain.
3. Why can the center of mass of an object lie outside the physical material of the object? Give two examples.
4. Explain the two factors that determine stability. For each one, explain why it improves stability using the concept of tipping torque.
5. What is the base of support? Explain carefully why the stability condition depends on the relationship between the center of mass projection and the base of support.
6. A tall bookshelf loaded only on the top shelf is less stable than one loaded uniformly. Explain using center of mass reasoning.
7. Explain why a sprinter at the start of a race leans forward so far that they are on the verge of tipping over. What role does instability play in their acceleration?
8. Explain the experimental procedure for locating the center of mass of an irregular flat object by hanging it from two different points.

## Center of Mass Problems

9. Two masses are placed on a light rod: a 3.0 kg mass at position x = 0 and a 5.0 kg mass at x = 0.80 m. Find the center of mass of the system.
10. Three masses are located as follows: 4.0 kg at (0, 0), 2.0 kg at (3.0, 0), and 6.0 kg at (1.5, 4.0). All positions in meters. Find the center of mass coordinates (x_cm, y_cm).
11. A uniform rod of mass 2.0 kg and length 1.2 m has a 5.0 kg mass attached to one end. Where is the center of mass of the system?
12. A non-uniform plank has its center of mass located 0.40 m from one end. The plank is 1.8 m long and has a mass of 6.0 kg. It rests on two supports placed at each end. What is the upward force exerted by each support?
13. A uniform L-shaped frame is made of two identical 1.0 kg rods, each 0.60 m long, arranged at right angles (one horizontal, one vertical from the right end of the first). Find the center of mass of the frame.

## Stability and Tipping Exercises

14. A rectangular box is 0.40 m wide and 0.90 m tall. Assuming uniform density, what is the maximum angle it can tilt from vertical before tipping over?
15. A uniform cube of side 0.50 m is placed on an inclined surface. At what angle of inclination does the cube begin to tip over? Assume the surface has enough friction to prevent sliding.
16. A vehicle has a center of mass 0.70 m above the ground and a track width (distance between wheels) of 1.60 m. What is the maximum sideways tilt angle before rollover?
17. A 60 kg person stands with feet together (base width approximately 0.10 m front-to-back) and center of mass at height 1.0 m. If they lean forward, at what forward lean angle does their center of mass reach the tipping point?

## Applied Engineering Stability Problems

18. A crane has its boom extending 8.0 m to one side of the central column. The load lifted is 3000 N. A counterweight is placed 3.0 m on the opposite side of the column. What must the counterweight's weight be to balance the crane's rotational tendency about the base?

19. A retaining wall is 3.0 m tall and 1.0 m wide at its base, made of concrete (density 2400 kg/m³, treated as uniform). The horizontal soil pressure acts as a force of 8000 N applied at a height of 1.0 m from the base. (a) Find the torque trying to overturn the wall about its toe. (b) Find the restoring torque due to the wall's own weight. (c) Is the wall stable?
20. A filing cabinet is 1.60 m tall and 0.50 m deep. Its center of mass is at 0.80 m height when empty. A fully loaded top drawer shifts the center of mass to 1.10 m height. Calculate the maximum tipping angle for each case and comment on the safety implications.

## Biomechanics Applications

21. A person bends forward so that their torso is horizontal. Their upper body (mass 40 kg) now has its center of mass 0.35 m in front of the hip joint. The back extensor muscles, attached 0.05 m behind the spine (pivot), must provide an upward force. Calculate the required muscle force.
22. A rock climber stands on a small foothold with feet together. Their center of mass is 1.0 m above the foothold and 0.15 m to the right of it. Describe the torque situation. What must the climber do to restore balance?
23. Explain, using center of mass and base of support concepts, why a person carrying a heavy briefcase in one hand instinctively leans to the opposite side.

## Chapter 21 Checkpoint

Before moving on to Chapter 22, confirm you can do each of the following without notes.

| I can… | See Section |
|---|---|
| Define center of mass and explain what it represents physically | **21.1** |
| Distinguish center of mass from geometric center and give an example of each | **21.2** |
| Calculate center of mass position for a system of discrete masses in 1D and 2D | **21.3** |
| Explain the relationship between center of gravity and center of mass near Earth's surface | **21.4** |
| Explain why the height of the center of mass affects stability | **21.5** |
| Define base of support and state the stability condition in terms of it | **21.6** |

| | |
|---|---|
| Describe the tipping point and calculate the critical tipping angle for a simple shape | **21.7** |
| Explain how center of mass shifts in a moving or flexible body | **21.8** |
| Apply stability principles to engineering examples such as cranes and retaining walls | **21.9** |
| Describe how athletes use center of mass management in their sport | **21.10** |
| Explain why a hanging object's center of mass lies below the suspension point | **21.11** |
| Identify and correct common errors in stability analysis | **21.12** |

*Continue to Chapter 22: Stress, Strain, and Elasticity →*